

VINELAND STATION RCS, ON, Canada

WMO: 711710

Lat: 43.191N

Lon: 79.399W

Elev: 79

StdP: 100.38

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -15.6	(c) -13.2	(d) -20.8	(e) 0.6	(f) -14.6	(g) -18.6	(h) 0.7	(i) -12.5	(j) 11.8	(k) -4.2	(l) 10.7	(m) -5.1	(n) 4.8	(o) 240	(p) 0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.8	(c) 31.2	(d) 22.4	(e) 29.6	(f) 21.7	(g) 28.0	(h) 21.0	(i) 24.0	(j) 29.0	(k) 23.0	(l) 27.6	(m) 22.2	(n) 26.2	(o) 4.9	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.2	26.7	21.5	16.3	25.8	20.7	15.5	25.0	72.3	29.1	68.7	27.6	65.6	26.4	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.0	(b) 8.8	(c) 7.9	(e) -18.3	(f) 33.5	(g) 2.9	(h) 1.6	(i) -20.3	(j) 34.7	(k) -22.0	(l) 35.6	(m) -23.6	(n) 36.5	(o) -25.7	(p) 37.7		
(a) 10.0	(b) 8.8	(c) 7.9	(e) -18.8	(f) 25.8	(g) 2.7	(h) 1.3	(i) -20.7	(j) 26.7	(k) -22.3	(l) 27.5	(m) -23.9	(n) 28.2	(o) -25.8	(p) 29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	-3.4	-2.9	1.3	7.0	13.0	18.9	22.3	21.5	18.0	11.4	5.3	0.2	(6)
(7)		DBStd	10.07	5.93	5.32	5.06	4.46	4.59	3.97	3.04	2.87	3.69	4.38	4.36	4.50	(7)
(8)		HDD10.0	1670	416	362	272	113	17	0	0	0	0	35	151	305	(8)
(9)		HDD18.3	3587	673	595	528	342	177	41	3	5	50	221	391	563	(9)
(10)		CDD10.0	1474	1	0	3	22	112	267	383	358	241	77	10	1	(10)
(11)		CDD18.3	348	0	0	0	1	13	58	127	104	41	5	0	0	(11)
(12)		CDH23.3	2347	0	0	0	12	119	424	907	622	239	24	0	0	(12)
(13)		CDH26.7	639	0	0	0	1	30	132	269	148	58	2	0	0	(13)
(14)	Wind	WSAvg	3.6	4.5	4.4	4.1	3.8	3.0	2.9	2.8	3.0	3.6	3.9	4.2	(14)	
(15)	Precipitation	PrecAvg	858	71	55	63	81	73	75	73	79	78	70	70	(15)	
(16)		PrecMax	1041	121	120	107	133	155	195	190	129	195	164	131	150	(16)
(17)		PrecMin	654	39	27	18	36	21	20	14	26	28	32	23	31	(17)
(18)		PrecStd	115	24	23	27	28	35	40	39	26	36	34	27	28	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.5	13.6	17.4	25.0	29.8	32.3	33.6	32.2	31.1	26.0	19.0	13.9	(19)
(20)			MCWB	11.9	9.7	12.7	16.4	21.4	22.1	23.4	23.4	22.3	19.9	14.2	11.3	(20)
(21)		2%	DB	10.4	9.5	14.1	20.8	26.8	30.3	31.4	29.9	28.4	23.0	16.5	10.8	(21)
(22)			MCWB	8.2	7.1	10.3	14.0	19.2	21.4	22.8	22.2	21.4	17.8	12.5	8.4	(22)
(23)		5%	DB	7.2	5.8	11.3	17.5	23.7	27.9	29.7	28.2	26.0	20.3	14.4	8.1	(23)
(24)			MCWB	4.9	3.3	7.8	12.1	17.1	20.7	21.8	21.5	20.4	15.9	10.7	5.9	(24)
(25)		10%	DB	4.3	3.6	8.0	14.0	20.9	25.6	27.8	26.7	24.1	18.1	12.2	6.1	(25)
(26)			MCWB	2.7	1.8	5.6	9.7	15.8	19.6	21.0	20.8	19.7	14.8	9.3	4.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	11.0	13.4	17.9	21.9	23.9	25.7	24.9	23.8	21.0	15.2	11.8	(27)
(28)			MCD B	14.4	13.0	16.9	23.9	28.3	29.9	30.9	29.5	28.6	25.0	17.5	13.4	(28)
(29)		2%	WB	8.5	7.0	10.8	15.1	20.1	22.6	24.2	23.6	22.5	18.7	13.0	8.8	(29)
(30)			MCD B	10.1	9.8	13.0	19.6	26.1	28.2	29.2	28.2	26.7	22.3	15.7	10.6	(30)
(31)		5%	WB	5.2	3.7	8.1	12.8	18.3	21.5	23.2	22.7	21.3	16.6	11.0	6.3	(31)
(32)			MCD B	6.8	5.3	11.0	17.3	22.5	26.4	27.6	26.6	24.8	19.2	13.7	8.0	(32)
(33)		10%	WB	2.7	1.8	5.5	9.8	16.4	20.3	22.2	21.9	20.2	15.2	9.4	4.4	(33)
(34)			MCD B	4.2	3.4	7.7	13.3	20.0	24.6	26.1	25.3	23.5	17.7	12.0	5.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.5	6.7	7.1	8.3	9.4	8.9	8.8	8.4	8.5	7.4	7.0	5.5	(35)
(36)			MCD BR	9.0	10.2	11.8	15.0	13.8	11.9	10.7	10.2	10.7	10.5	10.8	8.3	(36)
(37)			MCWBR	8.2	8.7	8.6	9.5	7.9	6.0	5.1	4.8	5.7	6.8	8.1	7.4	(37)
(38)		5% WB	MCD BR	8.6	9.8	11.2	14.0	13.3	10.9	9.0	8.6	9.3	8.9	9.8	8.2	(38)
(39)			MCWBR	8.1	8.7	8.7	10.0	8.4	6.3	5.0	4.7	5.5	6.5	8.0	7.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.300	0.332	0.341	0.385	0.412	0.444	0.441	0.425	0.390	0.368	0.318	0.303	(40)	
(41)		taud	2.424	2.320	2.389	2.301	2.276	2.220	2.253	2.284	2.360	2.394	2.501	2.462	(41)	
(42)		Ebn at Noon	836	858	900	881	862	833	831	831	836	808	810	798	(42)	
(43)		Edh at Noon	76	99	105	124	131	139	133	125	108	92	70	67	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.37	2.13	3.34	4.42	5.40	5.87	5.89	5.20	4.15	2.51	1.57	1.09	(44)	
(45)		RadStd	0.12	0.19	0.36	0.47	0.57	0.43	0.44	0.27	0.38	0.23	0.21	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47) Regional (52 neighbors)		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A

Nomenclature: See separate page